SQL PROJECT 4
WORKFORCE DATA ANALYSIS

Salary and Remote Work Trends Across Job Titles and Company Locations

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Platform: Microsoft SQL Server (T-SQL)

1. Introduction

This project analyses workforce salary patterns, job titles, experience levels, employment types, company locations, company sizes and remote-work arrangements. The goal is to support salary benchmarking, workforce planning, recruitment, retention and remote-work policy decisions.

2. Objectives

- Compare average salaries by job title, experience level and location.
- Examine salary growth between years.
- Measure full-time/part-time and fully remote work patterns.
- Identify countries and locations with high salary benchmarks or large-company representation.
- Demonstrate SQL access-control concepts and use salary evidence to explore career options.

3. Dataset and fields

The assignment describes a workforce dataset with: work_year, experience_level, employment_type, job_title, salary, salary_currency, salary_in_usd, employee_residence, remote_ratio, company_location and company_size. Experience codes include EN (entry-level), MI (mid-level), SE (senior-level) and EX (executive-level). Employment codes include FT (full-time) and PT (part-time). Remote ratio 0 means fully in-office; 100 means fully remote; intermediate values indicate hybrid work. Company size uses S, M and L.

4. Methodology

The SQL solution uses filtering with WHERE, grouping and aggregation, HAVING for minimum group counts, common table expressions (CTEs), conditional aggregation, ranking functions, joins, date/year comparisons and percentage-growth formulas. Salaries are compared using salary_in_usd to make values comparable across currencies.

5. Key analysis questions

The SQL script answers the assignment's 20 tasks: company-size counts for 2021; top part-time job titles by average salary in 2023 with a record-count threshold; mid-level salary benchmarks; senior salary locations; salary growth by title; entry-level growth by country; remote-ratio updates; experience-level salary updates; highest-average-salary year by title; FT/PT percentages; remote manager locations; top countries by large-company records; percentage fully remote above $100,000; entry-level location benchmarks; highest-paying country per job title; sustained country salary growth; remote adoption comparison between 2021 and 2024; salary increases by level/title; role-based access; and salary-benchmark support for career switching.

6. Important assumptions and limitations

The supplied PDF describes the schema and tasks but does not include the complete dataset. Therefore, the report does not claim numerical findings or identify actual top countries/job titles. The SQL script assumes a table named dbo.WorkforceData with the fields listed in the assignment. Load the full dataset before running analysis. The PDF gives examples of salary-increase percentages but does not specify all rates; the script labels its example rates as assumptions and rolls the update back by default. The remote-work update is also rolled back by default.

7. Actionable recommendations

- Use salary_in_usd for cross-country comparisons.

- Check sample sizes before acting on high or low average salaries.
- Compare like-for-like experience levels and job titles.
- Validate year-on-year results against changes in dataset composition.
- Treat SQL UPDATE statements as demonstrations until business rules are approved.
- Apply least-privilege access controls and test views/permissions with appropriate database roles.

8. Conclusion

The project provides a repeatable SQL framework for workforce compensation and remote-work analysis. Once the full dataset is loaded, the queries can generate evidence for salary benchmarking, hiring-market selection, remote-work policy, talent retention and career-path discussions. Final numerical conclusions should be written only after the queries have been executed and their outputs reviewed.